

Alexandra Kashani Motlagh

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Education

The Ohio State University Columbus, OH
B.S. in Computer Science & Engineering — Computer Graphics and Game Design Aug 2019 – May 2025
Awards: Bill & Sue Lhota Endowed Scholarship (2023–2025) · Wening Memorial Scholarship (2024–2025)

Publications

Kashani Motlagh, A. (First Author), Mehta, S., Ahmad, I., Clark, A., Lei, H.
A Performance Comparison Between Two Speech-To-ASL Gesture-Projection Translation Implementations.
World Congress in Computer Science, Computer Engineering & Applied Computing, **Springer Nature**, 2024.

Research Experience

University of Arizona & University of Oulu Oulu, Finland
NSF IRES Undergraduate Researcher, Dept. of Systems & Industrial Engineering May 2024 – Aug 2024

- **IRES Track I: US–Finnish Research on Sustainable Evolution and Technical Debt Management in Cloud-Native Systems**
- Analyzed 390+ open-source repositories from the MSR Dataset to identify Infrastructure-as-Code usage patterns in Terraform and Ansible, supporting development of an LLM-based system for automated cloud infrastructure generation
- Built filter-parsing tools for the top 13 IaC services and interfaced with 7 LLM models for repository classification

University of Texas at Arlington Arlington, TX
NSF REU Undergraduate Researcher, Dept. of Computer Science & Engineering Jun 2023 – Aug 2023

- **Conducted research on communication-assistive technologies** under Dr. Ishfaq Ahmad and Dr. Mari F. Tietze
- Surveyed state-of-the-art speech-to-ASL translation methods and experimented with Natural Language Models to propose a pioneered solution for real-time ASL gesture projection
- Explored latency optimization of an ASL-Gloss stage to improve text-to-ASL translation accuracy; results published in Springer Nature

University of Dayton / Air Force Research Laboratory Dayton, OH
Research Intern (Mentors: Dr. Matthew Scherreik & Dr. Tim Anderson) Summers 2022, 2023, 2024

- Analyzed and mitigated issues in beam search for neural machine translation at large beam widths
- Innovated “Reject Option Beam Search” to improve translation quality metrics (BLEU/COMET) at large beam widths
- Pioneered end-to-end training for naturally constrained Reject Option Classification

Wright State University / Air Force Research Laboratory Dayton, OH
Undergraduate Research Intern (Mentor: Dr. Roman Ilin) Summers 2020, 2021

- Engineered a semi-automated system for temporal satellite imagery collection and classification (**ICCV 2021 Workshop**)
- Innovated an ensemble distillation method to enhance model performance on ambiguous instances

Projects

Infrastructure as Code in Micro-services Dataset Python, AI/ML 2024
Identified IaC tools across 390 GitHub repositories using custom filter-parsing tools for 13 IaC services; queried 7 LLM models for automated classification validation.

IttyBityCity Three.js, JavaScript 2026
Built an interactive, explorable 3D city in the browser using Three.js with procedural generation and camera controls.

RushRoute JavaScript, HTML Canvas 2024
Created a fast-paced obstacle avoidance game with procedural level generation and increasing difficulty.

Technical Skills

Languages: Python, JavaScript, TypeScript, C#, SQL, HTML, CSS
Technologies: Three.js, WebGL, OpenGL, Git, Visual Studio, AI/ML Model Development
Research: Natural Language Processing, Computer Vision, Satellite Imagery Classification, LLM Integration

Awards & Organizations

Society of Women Engineers 2021 – 2025
Bill & Sue Lhota Endowed Scholarship, College of Engineering, Ohio State University 2023 – 2025
Wening Memorial Scholarship, College of Engineering, Ohio State University 2024 – 2025